Episodic Memory Without Learning: Ternary Neural Computation on Peripheral Hardware at 30 Microamps

Tripp Josserand-Austin
EntroMorphic Research
tripp@entromorphic.com

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Abstract

We present a three-layer ternary neural computing system on a $0.50 microcontroller that classifies wireless signals at 100% label-free accuracy in peripheral hardware at 430 Hz and accumulates a temporal model of what it has perceived—pattern-specific internal representations that develop from episodic memory retrieval over 120 seconds of live operation. The system draws approximately 30 microamps in autonomous mode. No floating point. No multiplication. No backpropagation. No training. Signatures enrolled from a 30-second observation window (sign of mean). CfC weights random and never updated.

The temporal model emerges from a 64-node Navigable Small World graph (2 KB of LP SRAM) that stores and retrieves episodic snapshots of the system's perceptual state. The retrieval-and-blend mechanism alone—without any gate bias or weight learning—produces 8.5–9.7/80 MTFP divergence across 4 wireless signal patterns. The episodic memory is causally necessary: ablation (CfC without VDB blend) collapses pattern pairs that VDB feedback separates.

We also report two honest negative results from mechanisms designed to improve on the VDB baseline: (1) agreement-weighted gate bias (kinetic attention) is consistently harmful at MTFP resolution (mean $-5.5/80$, 3 runs), and (2) Hebbian weight learning produces no improvement ($+0.1 \pm 1.1/80$ MTFP at $n = 3$).

The system's power is in its episodic memory, not in learned weights or attentional bias. Classification is structurally separated from the temporal model by a zero-weight wall ($W_f[\text{hidden}] = 0$) that holds across all experiments. All results are from bare silicon.

1 Introduction

The dominant paradigm for neural computation on microcontrollers treats the device as a constrained deployment target—a model trained elsewhere, quantized, and loaded onto the chip to perform inference. The computation is designed in floating point and adapted to the hardware. The hardware is an inconvenience to be managed.

This paper describes a system designed the other way. The computation emerged from the hardware's native capabilities—ternary arithmetic on peripheral routing fabric—and the architecture grew from asking what structure those capabilities naturally support. The result is not a compressed version of a larger model. It is a system that could not exist in any other substrate, because its computational primitives (DMA descriptor chains, pulse counters, parallel I/O loopback) have no floating-point analog.

The system has three layers:

Layer 1 (GIE): A circular DMA chain streams ternary-encoded weight-input products through a parallel I/O peripheral configured in loopback mode. Two pulse counter units accumulate agreement and

disagreement edges. An ISR decodes per-neuron dot products from cumulative differencing, applies a ternary CfC blend (UPDATE/HOLD/INVERT), re-encodes the updated hidden state into the DMA descriptors, and re-arms the parallel I/O byte counter. The chain loops at 430 Hz. After initialization, the CPU computes zero dot products.

Layer 2 (LP core): A 16 MHz RISC-V ultra-low-power core, drawing approximately 30 microamps, runs a second ternary CfC network in hand-written assembly. It reads the GIE's hidden state, computes 32 ternary intersections against its own weight matrices, and maintains a 16-trit hidden state. A 64-node Navigable Small World graph provides episodic memory—storing 48-trit snapshots of the system's perceptual state and retrieving the nearest match for feedback blending into the LP hidden state.

Layer 3 (HP core): The 160 MHz application core handles initialization, weight loading, ESP-NOW wireless reception, classification arbitration, and test orchestration. It is awake only when needed.

The operations used throughout are AND, popcount (byte lookup table), add, subtract, negate, branch, and shift. The operations absent are multiply, divide, floating point, gradients, and backpropagation.

1.1 The Claims

This paper demonstrates three things and honestly reports two failures:
Demonstrated:

1. Peripheral-hardware ternary dot products at 430 Hz classify 4 wireless signal patterns at 100% label-free accuracy (TEST 11).

2. A 64-node episodic memory layer produces pattern-discriminative temporal states at 8.5–9.7/80 MTFP divergence from VDB feedback alone (TEST 12).

3. The episodic memory is causally necessary—ablation collapses what VDB feedback separates (TEST 13).

Failed:

4. Agreement-weighted gate bias (kinetic attention) degrades LP representation at MTFP resolution: $-5.5/80$ mean across 3 runs (TEST 14).

5. Hebbian LP weight learning produces no improvement: $+0.1 \pm 1.1/80$ at $n = 3$ (TEST 15).

The VDB baseline IS the finding.

2 Architecture

2.1 The Geometry Intersection Engine

The GIE exploits a property of the ESP32-C6's peripheral fabric: the General DMA controller (GDMA) can stream data through the Parallel I/O transmitter (PARLIO) configured in 2-bit loopback mode, producing edge transitions on GPIO pins that are counted by two Pulse Counter (PCNT) units. By encoding ternary weight-input products as 2-bit pairs (positive = `01`, negative = `10`, zero = `00`), the PCNT accumulates agreement and disagreement edges across each neuron's 160-trit buffer. The dot product is agreement − disagreement.

The DMA descriptor chain is circular: 5 dummy neurons (all-zero buffers that absorb PCNT pipeline residue) followed by 64 real neurons (32 gate-pathway + 32 candidate-pathway), with the last descriptor pointing back to the first. Each neuron's buffer is followed by a 64-byte separator with EOF=1, which triggers a GDMA interrupt. The ISR captures cumulative PCNT values at each separator, decodes per-neuron dots via differencing, applies the CfC blend equation, re-encodes the hidden state into the DMA buffers, and re-arms PARLIO's byte counter.

The CfC blend equation in ternary:

$$
\begin{aligned}
f[n] &= \text{sign}(\text{dot}(W_f[n],\ [\text{input} \,|\, \text{hidden}])) \\
g[n] &= \text{sign}(\text{dot}(W_g[n],\ [\text{input} \,|\, \text{hidden}])) \\
h_{\text{new}} &= \begin{cases} h_{\text{old}} & f = 0 \quad \text{(HOLD)} \\ g & f = +1 \quad \text{(UPDATE)} \\ -g & f = -1 \quad \text{(INVERT)} \end{cases}
\end{aligned}
$$

The three blend modes create non-gradient dynamics: HOLD provides inertia, UPDATE provides responsiveness, and INVERT creates oscillation and convergence resistance. Binary CfC has only UPDATE and HOLD. The third mode is a consequence of the ternary constraint.

2.2 Classification

Input encoding (128 trits): The ESP-NOW packet is encoded into 128 ternary trits:

- $[0..15]$ RSSI thermometer (16 trits)
- $[16..23]$ Pattern ID one-hot (8 trits): **masked** under `MASK_PATTERN_ID_INPUT=1`
- $[24..87]$ Payload bits (64 trits): 8 bytes $\times$ 8 bits, bit→trit
- $[88..103]$ MTFP21 gap history (16 trits): inter-packet timing
- $[104..127]$ Zeroed (24 trits)

Label-free operation: The `MASK_PATTERN_ID=1` flag zeros pattern_id trits in the enrollment signatures. The `MASK_PATTERN_ID_INPUT=1` flag zeros them in the runtime GIE input. Under both flags, no label information exists anywhere in the system. Classification relies entirely on payload content and inter-packet timing.

TriX classification (ISR, 430 Hz): Four ternary signature vectors are installed as W_f gate weights, with 8 neurons per pattern group. The ISR extracts per-group dot products and publishes the argmax. Accuracy: 32/32 = 100% label-free. The structural guarantee $W_f[\text{hidden}] = 0$ ensures classification is independent of the LP prior, gate bias, or temporal accumulation.

Prior P1-P2 confusion (resolved): With the original P2 payload, label-free accuracy was 71%—P2 shared 48/64 payload trits with P1. The P2 payload was redesigned to produce a cross-dot of 29/96 (30%), restoring 100% label-free accuracy. This made the test fair, not easy.

2.3 The LP Core: Geometric CfC + Episodic Memory

The LP core runs a second CfC network with 16 hidden neurons, taking the GIE's 32-trit hidden state concatenated with its own 16-trit hidden state as a 48-trit input. Weights are random ternary, generated once at initialization and never updated.

The LP core's VDB stores up to 64 snapshots of the system's perceptual state—48-trit vectors comprising [gie_hidden | lp_hidden]. A Navigable Small World graph ($M = 7$ neighbors) provides sub-linear approximate nearest-neighbor search. On each classification event (CMD 5), the LP core runs one CfC step, searches the VDB, and blends the best match's LP-hidden portion into the current LP hidden state.

The blend rule:

- Agreement ($h = \text{mem}$): no change

- Gap fill ($h = 0$, mem $\neq 0$): adopt memory value
- Conflict ($h \neq 0$, mem $\neq 0$, $h \neq$ mem): set to 0 (HOLD)

The HOLD-on-conflict rule is the damper. A neuron in conflict reverts to the neutral state. This bounds the feedback loop—ternary values can change by at most 1 per step, and conflicting feedback produces inertia rather than oscillation.

2.4 The MTFP Measurement Resolution

The LP CfC computes 16 integer dot products per step (`lp_dots_f[16]`). The sign-quantized version (`lp_hidden[16]` = sign(`lp_dots_f`)) is the 16-trit vector used by the LP CfC's recurrence and the VDB. The MTFP-encoded version (5 trits per dot: sign + 2 exponent + 2 mantissa) is an 80-trit vector that preserves magnitude information.

The distinction is critical. Sign-space Hamming and MTFP Hamming can give **opposite results** for the same data (see Section 4.3). All divergence measurements in this paper use MTFP unless otherwise noted.

2.5 Memory Budget: 16 KB

The LP core's 16 KB SRAM contains:

- Vector table: 128 bytes
- Code (`.text`): ∼7,600 bytes
- Popcount LUT (`.rodata`): 288 bytes
- CfC state (`.bss`): ∼968 bytes
- VDB nodes (`.bss`): 2,048 bytes (64 × 32 bytes)
- VDB metadata: ∼80 bytes
- Free for stack: ∼4,400 bytes (peak: 608 bytes during VDB search)

Every byte is specified. Every instruction is hand-written. No compiler decisions, no hidden register allocation, no spills.

3 Results: What the Hardware Demonstrates

3.1 Classification: 100% Label-Free (TEST 11)

32/32 correct classifications. Both ISR TriX (430 Hz hardware) and CPU core_pred agree at 100%. Four wireless signal patterns with distinct payloads and timing characteristics, classified from a 30-second enrollment window using sign-of-mean signatures. No labels in the input, no labels in the signatures, no labels anywhere in the system.

This is not a compressed model running inference. It is an enrollment-based classifier where the enrollment happens on the same hardware that runs the classification, using the same ternary operations, in 30 seconds of live observation.

	P0	P1	P2	P3
P0	0	5	6	8
P1	5	0	9	13
P2	6	9	0	10
P3	8	13	10	0

Table 1: LP MTFP divergence matrix (/80). Mean: 8.5/80 (full suite) to 9.7 ± 0.6/80 (3-rep measurement). All 6 pairs separated. P3 (ramp: incrementing payload at 10 Hz) is the most distinctive.

3.2 Temporal Context: 8.5–9.7/80 MTFP Divergence (TEST 12)

With VDB feedback (CMD 5), 120 seconds of live wireless input from 4 cycling patterns produces the following LP MTFP divergence matrix:

This is the LP CfC's 16 dot products, MTFP-encoded, accumulated over 120 seconds of episodic memory retrieval. The VDB stores snapshots of [gie_hidden | lp_hidden] and blends the nearest match back into the LP state. Over time, the LP state develops pattern-specific magnitude profiles—different patterns produce different dot product distributions, visible in the MTFP encoding.

No labels drive this. No training signal. No gradient. The temporal model emerges from the interaction between the CfC projection (random but consistent) and the VDB episodic retrieval (stores what actually happened, blends it back in).

3.3 Causal Necessity: VDB Feedback Required (TEST 13)

CMD 4 (CfC + VDB search, no feedback blend) vs CMD 5 (CfC + VDB + feedback blend):

- CMD 4 P1-P2 sign Hamming: 1
- CMD 5 P1-P2 sign Hamming: 2, MTFP Hamming: 9

VDB feedback contribution: +1 sign trit, +8 MTFP trits for the P1-P2 pair. The VDB is causally necessary.

4 Honest Negatives: What We Tried and What We Learned

4.1 Kinetic Attention: Harmful at MTFP Resolution

Mechanism: Agreement-weighted gate bias. When the LP prior agrees with the TriX prediction, lower the gate threshold for the predicted pattern's neuron group.

Hypothesis: Gate bias should amplify LP divergence by making the GIE more pattern-specific.

Run	14A (no bias)	14C (full bias)	Improvement
1	9.8	10.2	$+0.4$
2	15.5	8.5	$\mathbf{-7.0}$
3	15.5	5.7	$\mathbf{-9.8}$
Mean	13.6	8.1	$\mathbf{-5.5}$

Table 2: MTFP divergence (/80) under kinetic attention. 3 runs, label-free, Test 14.

Diagnosis: The bias lowers the gate threshold for one neuron group. More neurons fire. A saturated GIE hidden state is more uniform across patterns, making the LP input less discriminative. The bias improves GIE sensitivity but degrades GIE discriminability.

The mechanism fires (per-group fire rate shift >10% every run). The effect is negative. Reported honestly.

4.2 Hebbian Learning: Noise at 16 Neurons

Three iterations, each addressing a limitation of the previous:

1. **VDB mismatch, f-only:** $+2.5$ with label in input, -1.7 genuinely label-free. The VDB error signal was exploiting the pattern_id leaked through the GIE hidden state.

2. **TriX accumulator, f-only:** -1.0 label-free. Clean target but ∼50% of errors were in the g-pathway, making f-pathway flips counterproductive.

3. **TriX accumulator, diagnosed f+g:** $+0.1 \pm 1.1/80$ MTFP at $n = 3$. The diagnosis fixed the direction (harmful → neutral) but the mechanism produces no improvement.

Diagnosis: The 16-neuron LP with random ternary weights produces 16 integer dot products. Each dot is the sum of ∼29 non-zero ternary products. Flipping one weight changes one dot by ± 2—a tiny perturbation in a flat landscape with vast equivalence classes.

The VDB feedback blend, by contrast, directly injects episodic content into LP hidden. It doesn't need to find good weights—it bypasses the weights entirely.

4.3 The Sign-Space Metric Can Mislead

The kinetic attention finding was initially reported as positive ($+1.3/16$ in sign-space) before the MTFP numbers were examined.

Run 2, the most dramatic case:

- Sign-space 14A: 0.0/16 (all patterns → identical sign vector)
- Sign-space 14C: 4.2/16 (bias "broke the symmetry")
- MTFP 14A: 15.5/80 (patterns well-separated in magnitude)
- MTFP 14C: 8.5/80 (bias crushed the magnitude diversity)

The LP states had identical signs but different magnitudes. Sign-space saw them as identical (Hamming 0). MTFP saw them as well-separated (15.5/80). The bias shuffled the signs (creating sign-space divergence) while crushing the magnitudes (destroying MTFP divergence). A net information loss that sign-space reported as a gain.

The lesson: When sign-space and MTFP disagree, trust MTFP.

5 The Structural Wall

The classification guarantee $W_f[\text{hidden}] = 0$ was verified across every experiment in the April 9–12 session, including label-free operation, Hebbian weight learning (LP W_f and W_g modified, GIE W_f untouched), and multiple seeds.

TriX accuracy remains 100% label-free in every configuration. The structural wall—the guarantee that the classifier cannot be influenced by the temporal model—is the project's most robust finding.

This guarantee enables the honest negative results. We could test kinetic attention and Hebbian learning because the structural wall guaranteed that no matter how badly those mechanisms performed, they could not degrade the 100% classification accuracy. The classifier is immune to the prior.

6 Related Work

6.1 Ternary Neural Networks

Ternary weight networks [1, 2] quantize trained floating-point models to $\{-1, 0, +1\}$ for efficient inference. XNOR-Net [3] and subsequent binary/ternary quantization methods [4] treat ternary arithmetic as a compression technique—the model is designed in float, trained in float, and projected into low-precision weights for deployment. The computation is an approximation of the trained model.

The Reflex inverts this relationship. Its weights are generated ternary and never trained. There is no floating-point model being approximated. The ternary operations (AND, popcount, branch) are the native computation, not a quantized proxy for multiply-accumulate. The architectural consequences are different: ternary CfC has three blend modes (UPDATE/HOLD/INVERT) where binary has two, producing non-gradient dynamics that would not arise from quantizing a trained continuous model.

6.2 Complementary Learning Systems

McClelland, McNaughton, and O'Reilly [5] proposed that intelligent systems require complementary fast (hippocampal) and slow (neocortical) learning systems. Kumaran et al. [6] updated the theory to emphasize replay-based consolidation. O'Reilly et al. [7] formalized the computational trade-offs between pattern separation (hippocampus) and pattern completion (neocortex).

The Reflex implements CLS with fixed neocortical weights—the CfC projection is random and never updated. This is closer to Benna and Fusi's [8] analysis of memory capacity in synaptic systems with constrained plasticity, where fixed random projections support episodic retrieval when the memory store has sufficient capacity. In our system, 64 NSW nodes in 2 KB suffice for 4 patterns; the capacity bound is an open empirical question (see Section 7).

The VDB feedback blend implements memory-based prediction that parallels Lengyel and Dayan's [9] account of hippocampal contributions to decision-making: stored episodes directly inform current state estimation, bypassing learned statistical summaries. Our finding that Hebbian consolidation produces no improvement ($+0.1 \pm 1.1$) is consistent with their prediction that episodic systems outperform statistical learners when episodes are few and representations are low-dimensional.

See companion paper (Stratum 2) for the full CLS analysis, including the transition experiment and the permanent-hippocampus reframe.

6.3 Neuromorphic and On-Device Computing

Hardware neural networks on dedicated neuromorphic chips—Intel Loihi [10], BrainChip Akida [11], and IBM TrueNorth [12]—implement spiking neural networks with on-chip learning rules (typically STDP). These systems achieve remarkable energy efficiency but require custom silicon. The Reflex uses commodity peripheral fabric (DMA, pulse counters, parallel I/O) on a $0.50 general-purpose microcontroller, repurposing peripherals designed for motor control and communication as a neural substrate.

In the on-device inference space, TinyML frameworks [13, 14] deploy quantized models (typically INT8) on microcontrollers using optimized runtimes (TFLite Micro, CMSIS-NN). These systems achieve

inference but not temporal modeling—they classify each input independently, without accumulating a representation of what has been perceived over time. The Reflex's VDB episodic memory layer adds this temporal dimension at $\sim$30 μA, below the sleep-mode power of most TinyML deployments that duty-cycle a full inference engine.

Approximate nearest-neighbor search on microcontrollers has been explored for resource-constrained retrieval [15], but typically in service of classification or recommendation. The NSW graph in our system serves a different purpose: it provides episodic memory that feeds back into the neural dynamics, creating a recurrent loop between perception and memory that operates without gradient-based learning.

6.4 Prior-Signal Separation

The structural wall ($W_f[\text{hidden}] = 0$) that separates classification from temporal modeling relates to recent work on hallucination mitigation in language models. Retrieval-augmented generation [16] provides external evidence but lacks architectural guarantees that the evidence cannot be overridden by the model's prior. See companion paper (Stratum 3) for the full five-component analysis of structural hallucination resistance.

7 Limitations

1. **Single seed for most measurements.** The full-suite label-free dataset is one seed (0xCAFE1234). Kinetic attention was measured across 3 runs. Hebbian was replicated at $n = 3$.

2. **Four patterns, one sender.** Classification uses 4 patterns with distinct payloads from a single ESP-NOW sender.

3. **JTAG attached.** All runs use USB-JTAG for serial output and power. The "peripheral-autonomous" claim requires UART-only verification with direct current measurement. Pending.

4. **MTFP metric is nonlinear.** MTFP Hamming counts differing trits in a nonlinear encoding. The 8.5–9.7/80 baseline should be interpreted as a proxy for dot-product diversity, not a calibrated distance.

5. **Transition experiment: single seed label-free.** The P1→P2 transition experiment (TEST 14C) has been validated label-free on Seed A. Multi-seed label-free TEST 14C has not been performed.

6. **16 neurons may be a ceiling.** The Hebbian null result may be specific to 16 neurons with 48-trit input. Wider LP or MTFP-targeted learning might enable weight learning.

8 Conclusion

A ternary peripheral-fabric classifier at 430 Hz on a $0.50 microcontroller discriminates 4 wireless signal patterns at 100% label-free accuracy and builds a temporal context layer with 8.5–9.7/80 MTFP divergence—from episodic memory retrieval alone, without gate bias, without weight learning, without labels, without training, without floating point.

The system draws approximately 30 microamps. The classification is structurally separated from the temporal model by a zero-weight wall that holds across every experiment. The episodic memory (64 nodes, 2 KB, NSW graph, 100 Hz search) is causally necessary for LP divergence.

We tried to improve on the VDB baseline with two mechanisms: agreement-weighted gate bias and Hebbian LP weight learning. Gate bias is harmful—it saturates the GIE hidden state, reducing LP magnitude

diversity. Hebbian learning is noise—the 16-neuron LP with random ternary weights is too coarsely quantized for single-trit flips to find useful structure.

The system's power is in its episodic memory. The temporal model is in the memories, not in the weights. The hardware computes the dot products. The memories hold the context. The structural wall guarantees they cannot corrupt each other.

A Hardware Constants

Parameter	Value
Chip	ESP32-C6FH4 (QFN32) rev v0.2
GIE rate	430.8 Hz (peripheral fabric)
LP core	16 MHz RISC-V, $\sim$30 μA
LP CfC	16 neurons, 48-trit input, fixed random weights
VDB	64 nodes, NSW graph ($M = 7$), 48-trit vectors
LP MTFP	80 trits (5 per neuron: sign + 2 exp + 2 mant)
TriX	4 patterns, 8 neurons/pattern, 100% label-free
Structural wall	$W_f[n][\texttt{CFC_INPUT_DIM}:] = 0$ for all neurons
Build flags	`MASK_PATTERN_ID=1`, `MASK_PATTERN_ID_INPUT=1`

Table 3: System parameters.

B Operations Used

Gate pathway: AND, popcount (byte LUT), add, subtract, branch, shift. Candidate pathway: same operations, different weight matrix. Absent: multiply, divide, floating point, gradients, backpropagation.

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